

Shared document for the internship

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Link for the GitHub: <https://github.com/TheoBeille/Research-Intership/tree/main>

1 Finding the monotone inclusion

Our initial problem is:

$$\min_u \frac{1}{2} \|Ku - y\|_2^2 + \text{TGV}_{\alpha_0, \alpha_1}^2(u). \quad (1)$$

where

$$\text{TGV}_{\alpha_0, \alpha_1}^2(u) = \min_w \alpha_1 \|\nabla u - w\|_1 + \alpha_0 \|\mathcal{E}w\|_1,$$

with ∇ denoting the discrete gradient and $\mathcal{E}w = \frac{1}{2}(\nabla w + \nabla w^\top)$ is the symmetrized gradient. The full primal problem becomes

$$\min_{u, w} \frac{1}{2} \|Ku - y\|_2^2 + \alpha_1 \|\nabla u - w\|_1 + \alpha_0 \|\mathcal{E}w\|_1. \quad (2)$$

where $\|z\|_1 = \sup_{\|p\|_\infty \leq 1} \langle p, z \rangle$.

Relating it to the monotone inclusion problem We introduce dual variables:

$$p \text{ with } \|p\|_\infty \leq \alpha_1 \quad \text{and} \quad q \text{ with } \|q\|_\infty \leq \alpha_0.$$

This yields the saddle formulation of (2):

$$\min_{u, w} \max_{p, q} \frac{1}{2} \|Ku - y\|_2^2 + \langle p, \nabla u - w \rangle + \langle q, \mathcal{E}w \rangle - \iota_{\|p\|_\infty \leq \alpha_1}(p) - \iota_{\|q\|_\infty \leq \alpha_0}(q),$$

where ι_C denotes the indicator function of a convex set. The Lagrangian is

$$\mathcal{L}(u, w, p, q) = \frac{1}{2} \|Ku - y\|_2^2 + \langle p, \nabla u - w \rangle + \langle q, \mathcal{E}w \rangle - \iota_{\|p\|_\infty \leq \alpha_1}(p) - \iota_{\|q\|_\infty \leq \alpha_0}(q)$$

so the optimality condition

$$0 \in \partial \mathcal{L}(u, w, p, q).$$

can be explicitly written as

$$\begin{cases} 0 = K^\top(Ku - y) + \nabla^\top p, \\ 0 = -p + \mathcal{E}^\top q, \\ 0 \in \nabla u - w + \partial \iota_{\|p\|_\infty \leq \alpha_1}(p), \\ 0 \in \mathcal{E}w + \partial \iota_{\|q\|_\infty \leq \alpha_0}(q). \end{cases} \quad (3)$$

If we set $x = (u, w, p, q)$, then we can write the above optimality condition (3) as the following monotone inclusion problem:

$$0 \in Ax + Cx,$$

where

$$Cx = \begin{pmatrix} K^\top(Ku - y) \\ 0 \\ 0 \\ 0 \end{pmatrix} = \left(\nabla_u \frac{1}{2} \|Ku - y\|_2^2 \quad 0 \quad 0 \quad 0 \right).$$

and

$$A = \begin{pmatrix} 0 & B^\top \\ -B & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \partial \iota_{\|p\|_\infty \leq \alpha_1}(p) \\ \partial \iota_{\|q\|_\infty \leq \alpha_0}(q) \end{pmatrix} \quad \text{with} \quad B = \begin{pmatrix} \nabla & -I \\ 0 & \mathcal{E} \end{pmatrix}.$$

Then, we have

$$Ax = A \begin{pmatrix} u \\ w \\ p \\ q \end{pmatrix} = \begin{pmatrix} \nabla^\top p \\ -p + \mathcal{E}^\top q \\ -(\nabla u - w) \\ -\mathcal{E}w \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \partial \iota_{\|p\|_\infty \leq \alpha_1}(p) \\ \partial \iota_{\|q\|_\infty \leq \alpha_0}(q) \end{pmatrix}.$$

Function Spaces

The relevant spaces are:

- $X = L^2(\Omega)$
- $V = X \times X$: range of ∇ .
- $W = X \times X \times X \times X$: range of E .

Therefore, $\mathcal{X} = X \times V \times V \times W$ is the domain of A

2 Implementation of Algorithm 1

To recall

Algorithm 1 Forward-Backward Splitting with History and Deviations

1: **Input:** $x_0 \in H$, strictly positive sequences $(\gamma_n)_{n \in \mathbb{N}}$ and $(\lambda_n)_{n \in \mathbb{N}}$, non-negative sequences $(\zeta_n)_{n \in \mathbb{N}}$ and $(\mu_n)_{n \in \mathbb{N}}$, $\bar{\beta} \geq \beta \geq 0$

2: **Define** for all $n \in \mathbb{N}$:

$$\alpha_n := \frac{\mu_n}{\lambda_n + \mu_n}, \quad \bar{\alpha}_n := \frac{\gamma_n \mu_n}{\gamma_{n-1}(\lambda_n + \mu_n)}, \quad \theta_n := (4 - \gamma_n \bar{\beta})(\lambda_n + \mu_n) - 2\lambda_n^2$$

$$\hat{\theta}_n := 2\lambda_n + 2\mu_n - \gamma_n \bar{\beta} \lambda_n^2, \quad \bar{\theta}_n := \lambda_n + \mu_n - \lambda_n^2, \quad \tilde{\theta}_n := (\lambda_n + \mu_n) \gamma_n \bar{\beta}$$

3: **Set:** $y_{-1} = p_{-1} = z_{-1} = x_0$, $\Delta_0^1 = \Delta_0^2 = 0$, $\gamma_{-1} = \gamma_0$

4: **for** $n = 0, 1, 2, \dots$ **do**

5: $y_n = x_n + \alpha_n(y_{n-1} - x_n) + \Delta_n^1$

6: $z_n = x_n + \alpha_n(p_{n-1} - x_n) + \bar{\alpha}_n(z_{n-1} - p_{n-1}) + \frac{\bar{\theta}_n \gamma_n \bar{\beta}}{\hat{\theta}_n} \Delta_n^1 + \Delta_n^2$

7: $p_n = (I + \gamma_n A)^{-1}(z_n - \gamma_n C y_n)$

8: $x_{n+1} = x_n + \lambda_n(p_n - z_n) + \bar{\alpha}_n \lambda_n(z_{n-1} - p_{n-1})$

9: Choose Δ_{n+1}^1 and Δ_{n+1}^2 such that $(\lambda_{n+1} + \mu_{n+1}) \left(\frac{\tilde{\theta}_{n+1}}{\hat{\theta}_{n+1}} \|\Delta_{n+1}^1\|^2 + \frac{\hat{\theta}_{n+1}}{\theta_{n+1}} \|\Delta_{n+1}^2\|^2 \right) \leq \zeta_{n+1} l_n$

where l_n is defined as:

$$l_n = \frac{\theta_n}{2} \left\| p_n - x_n + \alpha_n(x_n - p_{n-1}) + \frac{\gamma_n \bar{\beta} \lambda_n^2}{\hat{\theta}_n} \Delta_n^1 - \frac{2\bar{\theta}_n}{\theta_n} \Delta_n^2 \right\|^2$$

$$+ 2\mu_n \gamma_n \left\langle \frac{z_n - p_n}{\gamma_n} - \frac{z_{n-1} - p_{n-1}}{\gamma_{n-1}}, p_n - p_{n-1} \right\rangle$$

$$+ \frac{\mu_n \gamma_n \bar{\beta}}{2} \|p_n - y_n - (p_{n-1} - y_{n-1})\|^2$$

10: **end for**
